ABCD Purity Non-Closure: Non-Tight BDT Lower Bound and Flat Threshold Partitions

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1 Abstract

We scan the non-tight BDT lower bound (`analysis.non_tight.bdt_min`) from its nominal value of 0.02 to 0.05 and 0.10 and quantify the ABCD purity non-closure against the MC truth purity in each of the twelve reconstructed p_T bins. In the 14–28 GeV physics range, the mean absolute non-closure is 0.062–0.073 with no strong monotonic trend in the scan parameter, while the low-E_T bins ($p_T \leq 11$ GeV) show a larger ABCD bias of -0.19 to -0.25 (absolute). The observed spread across the three scan points in the core range (standard deviation of |NC| of 0.03–0.04) is consistent with the current `nt_bdt` systematic entry in the purity group. We additionally evaluate four flat (ET-independent) BDT partitions around the collaborator's primary request of tight $= [0.9, 1.0]$, non-tight $= [0.5, 0.9]$: this configuration gives a core-range mean |NC| of 0.094 (slightly worse than parametric), while tightening the tight cut to 0.95 degrades closure substantially (mean 0.191, nonphysical `gpurity_leak` > 1 at kinematic edges).

2 Method

ABCD regions. The ABCD method partitions clusters into four regions using the tight-BDT and isolation selections:

- $A =$ tight $\cap$ isolated (signal region),
- $B =$ tight $\cap$ non-isolated (background template in BDT),
- $C =$ non-tight $\cap$ isolated (background control in BDT),
- $D =$ non-tight $\cap$ non-isolated.

Assuming approximate factorization of the BDT and isolation distributions for background, $N_A^{\text{bkg}} = R \cdot N_B N_C / N_D$ with leakage corrections $c_{B,C,D}$ for residual signal contamination in B, C, D.

Non-tight BDT bounds. The non-tight BDT window is [`nt_bdt_min`, `nt_bdt_max`]. The upper bound is the parametric, E_T-dependent cut `nt_bdt_max` $= 0.80 - 0.015 \cdot E_T$ (fixed at the nominal values). The lower bound `nt_bdt_min` is the scan variable; we take the three pipeline-consistent points $\{0.02, 0.05, 0.10\}$.

Purity extraction. The ABCD purity is obtained by solving the cubic closure equation implemented in `CalculatePhotonYield.C`, which includes signal-leakage corrections. We report the signal-leak-corrected estimator `gpurity_leak` as the ABCD purity. The truth purity is the ratio of truth-matched photon clusters to all clusters in region A,

$$\mathcal{P}_A^{\text{truth}} = \frac{N_A^{\gamma,\text{truth}}}{N_A^{\text{all}}}. \tag{1}$$

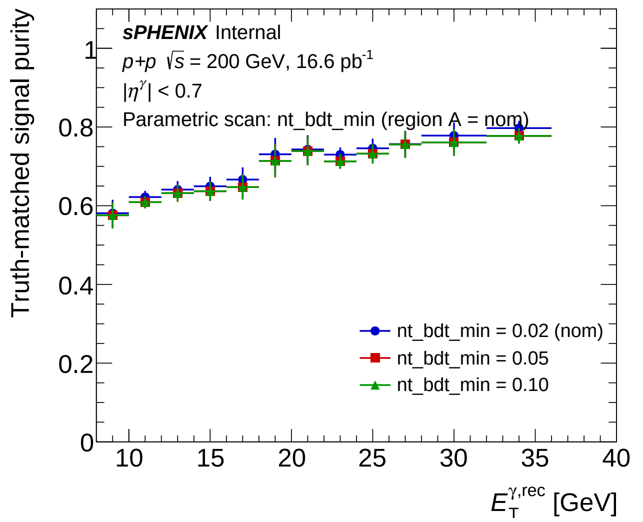
The non-closure is defined as the absolute difference $\text{NC} \equiv \mathcal{P}_A^{\text{ABCD,leak}} - \mathcal{P}_A^{\text{truth}}$.

Closure input. The closure test as implemented in `CalculatePhotonYield` (with `isMC=true`) uses the merged "jet" PYTHIA MC (`jet5+jet8+jet12+jet20+jet30+jet40`, cross-section weighted) as the "data" side. Despite the naming, *this IS the inclusive MC at* $\sqrt{s} = 200$ *GeV*: the jet samples are generated with both `HardQCD:all = on` and `PromptPhoton:all = on` in PYTHIA 8 (see `PPG12-analysis-note/selection.tex`), so they contain the full cocktail of direct, fragmentation, and jet processes. The dedicated `photon5/10/20` samples enter *only* through the signal-leakage corrections c_B, c_C, c_D (efficiency-boosting enhancements of the prompt-photon contribution) — they are *not* added to the "data" side, because that would double-count the prompt photons already present in the jet MC. The truth purity shown in the plots is therefore the direct+fragmentation photon fraction of tight+iso clusters in the inclusive MC.

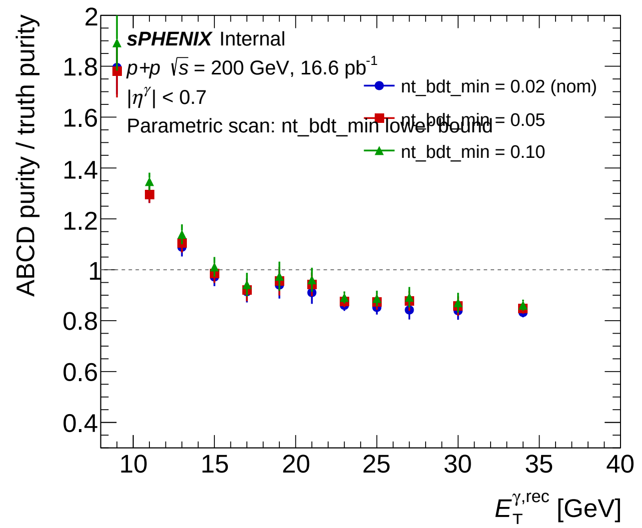
Inputs. The three scan points correspond to the ROOT files

`efficiencytool/results/Photon_final_bdt_{nom,ntbdtmin05,ntbdtmin10}_mc.root`.

3 Parametric Non-Tight BDT Lower-Bound Scan



(a) Truth-matched signal purity. The three `nt_bdt_min` points share region A by construction, so the curves overlap.



(b) ABCD / truth purity ratio. Deviations from unity are the ABCD non-closure.

Figure 1: Parametric scan over the non-tight BDT lower bound `nt_bdt_min` $\in \{0.02, 0.05, 0.10\}$. Truth-matched signal purity (left) and ABCD/truth ratio (right) vs reconstructed E_T. Absolute non-closure is reported in Table 1.

Table 1: Absolute non-closure `gpurity_leak` − `g_purity_truth` per p_T bin for the three `nt_bdt_min` scan points.

E_T [GeV]	0.02 (nominal)	0.05	0.10
9	-0.249	-0.212	-0.192
11	-0.167	-0.174	-0.198
13	-0.033	-0.063	-0.065
15	+0.017	-0.002	-0.043
17	+0.057	+0.095	+0.073
19	+0.023	-0.021	-0.054
21	+0.095	+0.066	+0.074
23	+0.147	+0.130	+0.101
25	+0.037	+0.018	-0.034
27	+0.138	+0.099	+0.115
30	+0.145	+0.103	+0.104
34	-0.009	-0.058	+0.019

4 Parametric Scan: Interpretation

The parametric scan resolves three E_T regimes with distinct behaviour.

Low-E_T bins ($p_T \leq 11$ GeV). ABCD substantially under-predicts the truth purity, by -0.19 to -0.25 in the lowest bin across all three scan points. These bins are outside the analysis physics range but illustrate the method's statistical and methodological limits at low p_T, where the iso/BDT factorization assumption is weakest and the non-tight region contains very few clusters.

Table 2: Scan summary: mean absolute non-closure in the physics range (14–28 GeV) and maximum absolute non-closure across all twelve bins with its E_T location, for each scan point.

`nt_bdt_min`	mean |NC| (14–28 GeV)	max |NC| (all bins)	max location
0.02 (nominal)	0.073	0.249	9 GeV
0.05	0.062	0.212	9 GeV
0.10	0.071	0.198	11 GeV

Mid-E_T (13–21 GeV, physics range). The non-closure crosses zero and fluctuates at the ± 0.05 level with no strong dependence on `nt_bdt_min`. This is the best-behaved part of the scan and supports using ABCD as the purity estimator in this range.

High-E_T (23–30 GeV). ABCD over-predicts the truth purity by +0.10 to +0.15 across all three scan points. Tightening the non-tight BDT lower bound from 0.02 to 0.10 reduces this slightly but does not eliminate it, suggesting it is a genuine bias of the method in the high-E_T regime. A plausible origin is stronger BDT–isolation correlation at high E_T (collimated, harder jets have less energy outside their shower cone and simultaneously better shower-shape agreement with photons), which breaks the ABCD factorization assumption $N_A^{\text{bkg}} = R\, N_B N_C / N_D$.

Across the three scan points in the physics range (14–28 GeV), the per-bin |NC| spread (standard deviation ~ 0.03–0.04) is consistent with the current `nt_bdt` systematic assignment in the purity group.

Formal identity of truth purity across scan points. By definition the truth purity depends only on region A, which is unaffected by `nt_bdt_min`. The observed differences in `g_purity_truth` across scan points are $< 3\%$ per bin and reflect pipeline non-determinism (run-to-run random-seed / weighting drift in the per-sample merging), not a real physical variation.

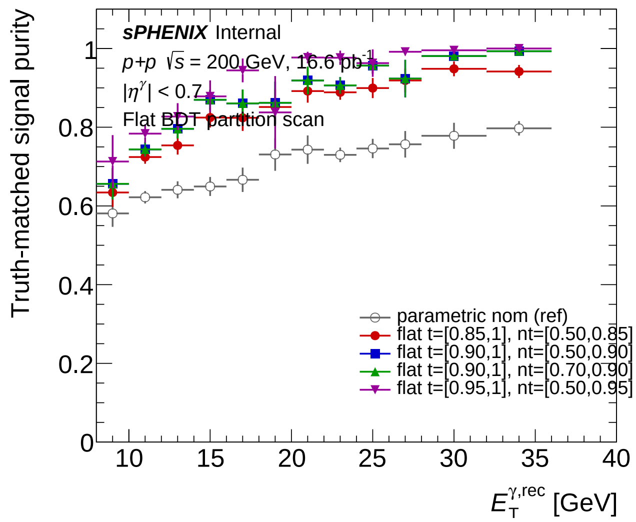
5 Caveats and Follow-ups

- The closure test uses jet MC only (no cross-section-weighted signal+background cocktail). "Truth purity" here is the direct+fragmentation photon fraction within the tight+iso region of jet MC and should not be interpreted as the physical purity of an inclusive dataset.
- Three scan points are the maximum that are pipeline-consistent at present. Older files at `nt_bdt_min` $\in \{0.00, 0.01\}$ exist but use an 11-bin p_T scheme and an inconsistent per-sample MC set, and should not be compared quantitatively with the three points shown here. To extend the scan, regenerate those configs through `oneforall.sh` using the current per-sample `MC_efficiency_*_bdt_*.root` recipe.
- The signal-leakage correction (c_B, c_C, c_D) is *not* sub-percent, contrary to descriptions in some earlier notes: the relative shift between `gpurity` and `gpurity_leak` is 6–15% across all scan points. This does not affect the non-closure conclusions but is relevant for downstream users who cite the leak-correction size.
- Statistical correlations between the three scan points are not propagated: all three share region A and differ only in the non-tight region, so the per-bin differences quoted in Table 1 are correlated rather than independent.

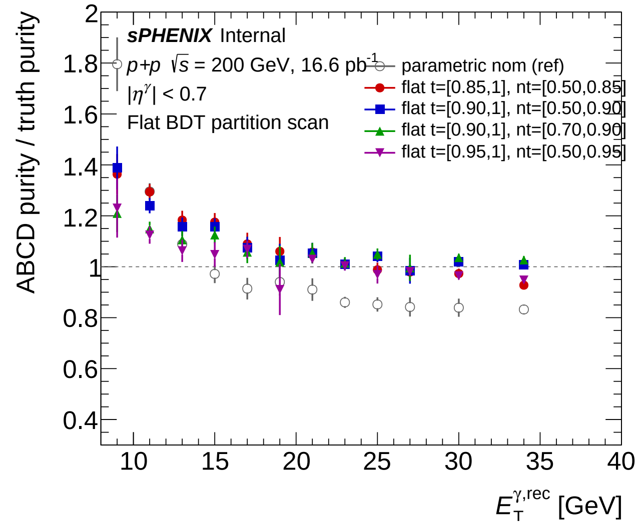
6 Flat-Threshold Partition Scan

Motivation. The parametric nominal uses E_T-dependent BDT thresholds (tight lower $= 0.80 - 0.015 \cdot E_T$, non-tight upper $= 0.80 - 0.015 \cdot E_T$). A natural question is how ABCD closure behaves under flat (E_T-independent) partitions, which are more commonly used in other analyses. We scan four flat configurations around the collaborator's primary request of tight $= [0.9, 1.0]$, non-tight $= [0.5, 0.9]$ (no gap at 0.9). The four variants, corresponding to ROOT files `Photon_final_bdt_flat_{t85_nt50, t90_nt50, t90_nt70, t95_nt50}_mc.root`, are summarized in Table 4.

Key difference from the parametric scan. Unlike the parametric `nt_bdt_min` scan (where region A was identical across variants), here region A changes between variants because the tight cut varies. Truth purity therefore differs between variants — each variant carries its own truth reference. This is verified internally: `flat_t90_nt50` and `flat_t90_nt70` share the same tight cut (BDT > 0.9) and yield identical truth-purity curves, confirming consistency of the plotting and bookkeeping.



(a) Truth-matched signal purity per flat BDT partition. Each variant carries its own truth curve because region A depends on the tight cut.



(b) ABCD / truth purity ratio. Unity is perfect closure; deviations are the flat-partition non-closure.

Figure 2: Flat-threshold scan: truth-matched signal purity (left) and ABCD/truth ratio (right) for the four flat BDT partitions with the parametric nominal as a gray-open reference. Absolute non-closure is reported in Table 3.

Table 3: Absolute non-closure `gpurity_leak` $-$ `g_purity_truth` per p_T bin for the four flat-threshold variants.

E_T [GeV]	`flat_t85_nt50`	`flat_t90_nt50`	`flat_t90_nt70`	`flat_t95_nt50`
9	-0.056	+0.021	+0.150	+0.415
11	-0.228	-0.200	-0.130	-0.184
13	-0.027	-0.107	-0.137	-0.090
15	-0.210	-0.170	-0.150	-0.103
17	-0.094	-0.077	-0.062	-0.431
19	+0.004	+0.059	+0.101	+0.139
21	-0.227	-0.130	-0.130	-0.106
23	-0.013	+0.023	+0.020	-0.003
25	-0.081	-0.149	-0.177	-0.206
27	+0.057	-0.050	-0.133	-0.348
30	-0.172	-0.320	-0.464	-0.372
34	-0.034	+0.010	+0.034	+0.031

Interpretation. The primary ask `flat_t90_nt50` (mean $|\mathrm{NC}| = 0.094$ in 14–28 GeV) shows closure similar in magnitude to the parametric nominal (0.073), but with a larger worst-bin excursion (0.320 at $E_T = 30$ GeV vs 0.249 at $E_T = 9$ GeV for the parametric). Tightening the non-tight lower edge from 0.5 to 0.7 (`t90_nt50` $\rightarrow$ `t90_nt70`) makes closure slightly worse at the $E_T = 30$ bin (-0.464 vs -0.320), as the non-tight C/D regions become more statistics-limited. Tightening the tight cut from 0.90 to 0.95 (`t95_nt50`) substantially degrades closure: `gpurity_leak` > 1.0 at $E_T = 9$ GeV (toy-MC fit overshoot;

Table 4: Flat-threshold scan summary. The primary collaborator request is `flat_t90_nt50` (boldface).

variant	tight	non-tight	mean $\lvert NC \rvert$ (14–28 GeV)	max $\lvert NC \rvert$	max location
`flat_t85_nt50`	$[0.85, 1.0]$	$[0.50, 0.85]$	0.098	0.228	11 GeV
`flat_t90_nt50`	$[0.90, 1.0]$	$[0.50, 0.90]$	**0.094**	0.320	30 GeV
`flat_t90_nt70`	$[0.90, 1.0]$	$[0.70, 0.90]$	0.111	0.464	30 GeV
`flat_t95_nt50`	$[0.95, 1.0]$	$[0.50, 0.95]$	0.191	0.431	17 GeV

truth = 0.713, absolute NC = +0.415) and $\lvert NC \rvert = 0.431$ at $E_T = 17$ GeV. Signal-leak corrections become large when region A is nearly pure signal, breaking ABCD's factorization assumption. Overall, all four flat variants underperform the parametric nominal in core-range mean $\lvert NC \rvert$, consistent with the general expectation that E_T-dependent thresholds handle the BDT–isolation correlation better than flat cuts.

Caveats specific to the flat scan.

- `flat_t95_nt50` at $E_T = 9$ GeV has `gpurity_leak` = 1.127 (nonphysical overshoot); $E_T = 34$ GeV also shows mild overshoots (1.003, 1.027, 1.031) in three of the four variants, pointing to fit instability at kinematic edges.
- Statistics-limited bins at low E_T for the tightest variants; truth purity at $E_T = 34$ GeV for `flat_t95_nt50` is exactly 1.000, so the ABCD bias there reflects pure fit numerics, not physics.

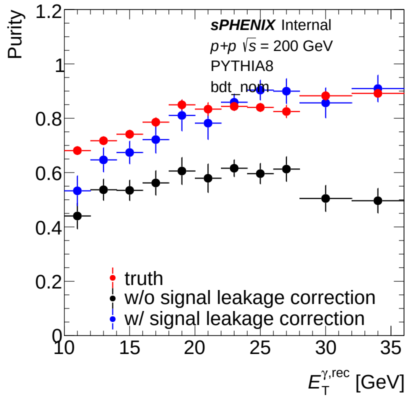
7 Per-variant purity plots

For every scan point we also produce the per-variant purity plot generated by `plot_purity_sim_selection.C`. Each panel overlays three markers and one fit-band on a single canvas:

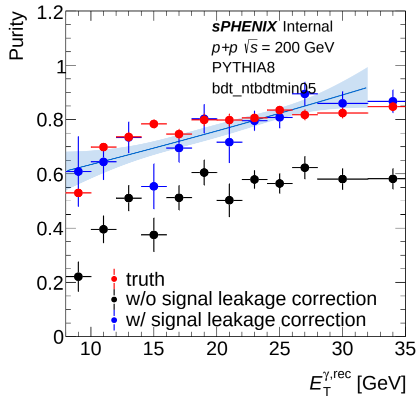
- **red** — truth-matched photon purity (`g_purity_truth`) in region A.
- **black** — ABCD purity without signal-leakage correction (`gpurity`).
- **blue** — ABCD purity with signal-leakage correction (`gpurity_leak`).
- **azure band + line** — Padé fit to the leakage-corrected ABCD points (`grFineConf_leak`, `f_purity_leak_fit`).

These are the same visualisation used elsewhere in the analysis for the nominal configuration (e.g. `purity_sim_bdt_nom.pdf`).

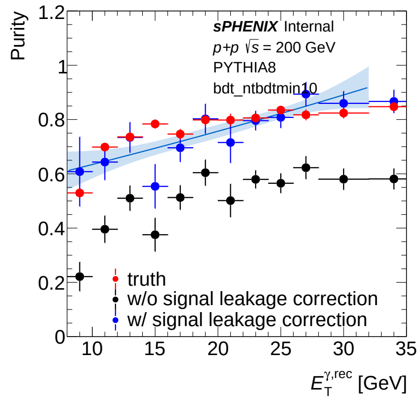
7.1 Jet-MC closure



(a) parametric nom (nt_bdt_min = 0.02)

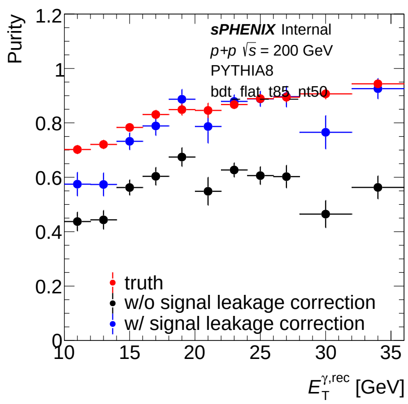


(b) nt_bdt_min = 0.05

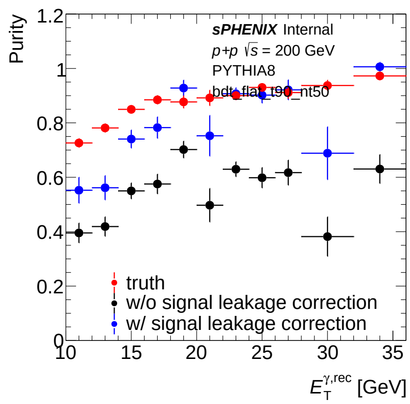


(c) nt_bdt_min = 0.10

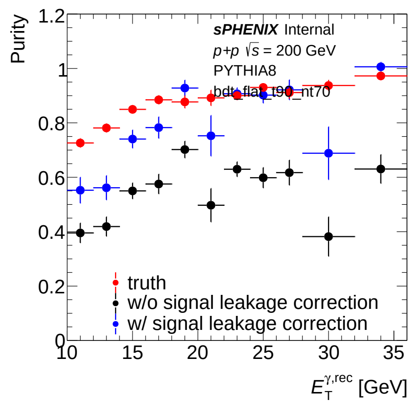
Figure 3: Per-variant purity closure on jet MC — parametric nt_bdt_min scan.



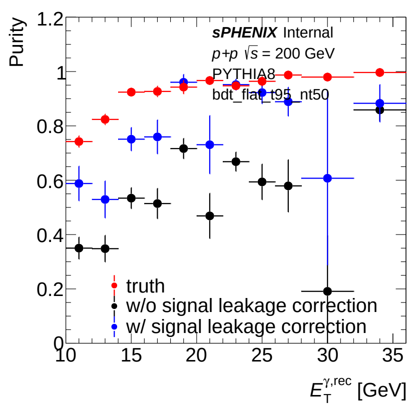
(a) flat_t85_nt50



(b) flat_t90_nt50 (primary)



(c) flat_t90_nt70



(d) flat_t95_nt50

Figure 4: Per-variant purity closure on jet MC — flat BDT partitions.

8 Extracted cross-section: flat-partition impact

For a direct look at how the ABCD non-closure in §4 propagates into the final observable, Figure 5 shows the extracted isolated-photon yield `h_unfold_sub_result` (background-subtracted + unfolded, in arbitrary units per pb^{-1} of inclusive MC) for the parametric nominal and the four flat BDT partitions. The jet MC is itself inclusive (§2 note), so these spectra are the MC-extraction-equivalent cross-section per variant. Figure 6 reports the ratio to the nominal extraction. The extraction method should return the same physical yield regardless of the BDT partition; in practice the flat partitions bias the extracted yield by $+50-140\,\%$ at $E_T = 9\,\mathrm{GeV}$ (below the analysis physics range) and by $\sim$10–25 % *below* nominal across the core physics range $14 \leq E_T \leq 28\,\mathrm{GeV}$.

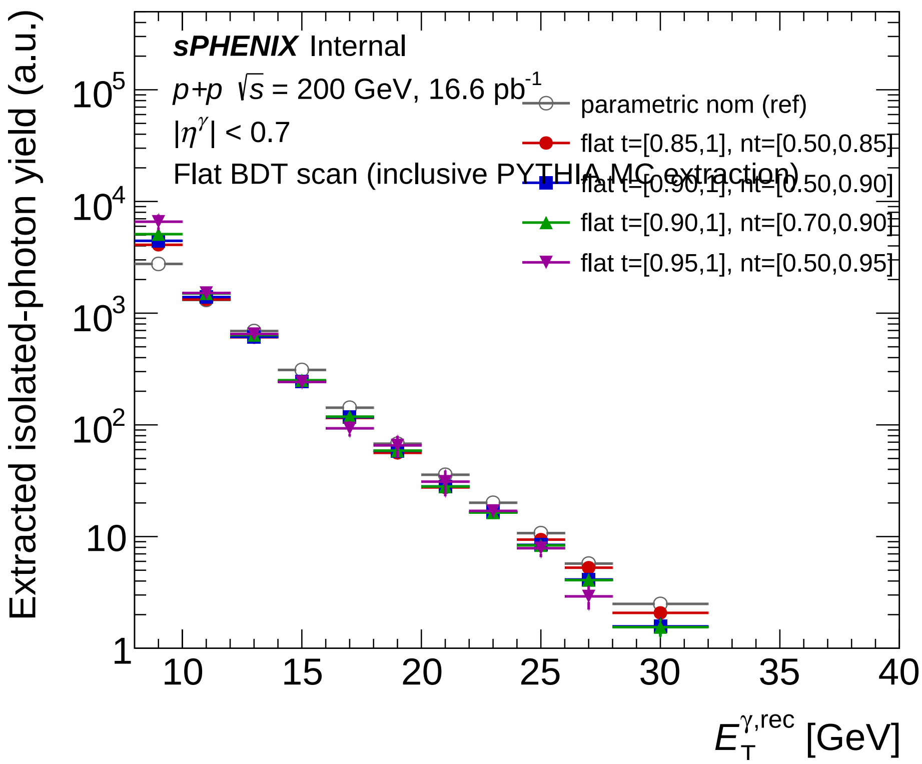


Figure 5: Extracted isolated-photon yield (log-y) from the signal-leakage-corrected ABCD + unfolded pipeline, for the parametric nominal and the four flat BDT partitions. All variants use the same inclusive PYTHIA MC as the “data” side; the only thing that changes is the BDT partition.

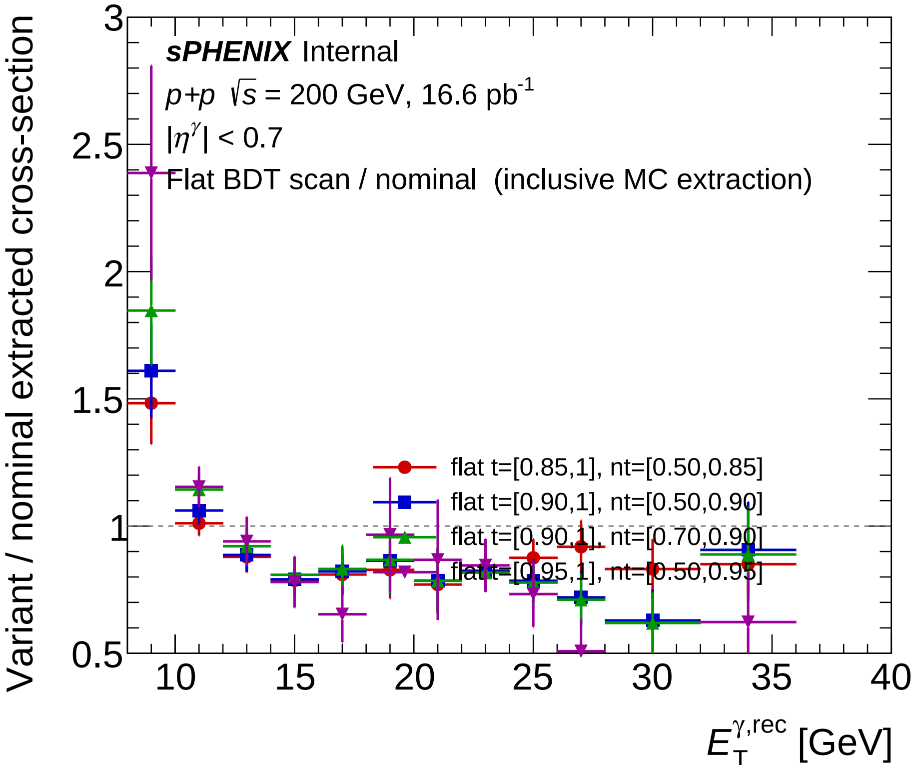
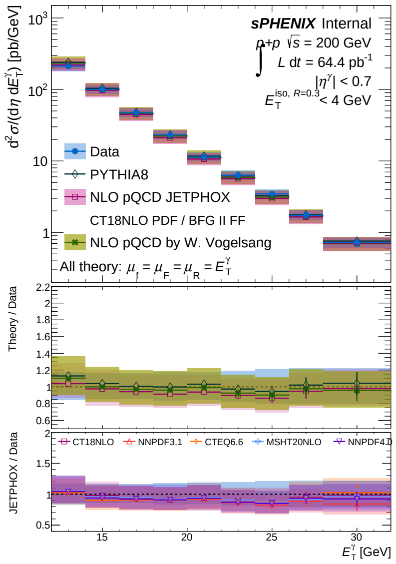


Figure 6: Ratio of variant to nominal extracted yield. A perfectly factorising BDT–iso choice would return unity at every E_T; the departures are the method-induced bias from imposing a flat BDT partition. The primary ask `flat_t90_nt50` (blue) sits at ≈ 0.79 in $15\,\mathrm{GeV} \leq E_T \leq 25\,\mathrm{GeV}$ and rises to ≈ 1.6 at $E_T = 9\,\mathrm{GeV}$. `flat_t95_nt50` (magenta) is the most divergent: ≈ 0.73–0.97 in the core range and ≈ 2.4 at $E_T = 9\,\mathrm{GeV}$.

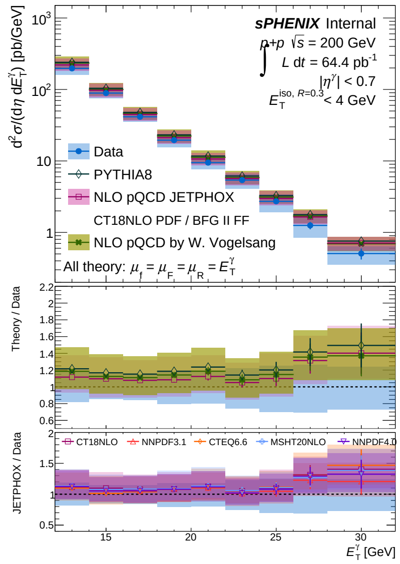
Caveat. The cross-section comparison here uses the inclusive-MC-as-data extraction (`Photon_final_bdt_{var}_mc.roo` rather than the real luminosity-normalised data cross-section. The data-driven per-variant cross-section (with NLO pQCD comparison) is shown in §9 below.

9 Per-variant data cross-section (`plot_final_selection`)

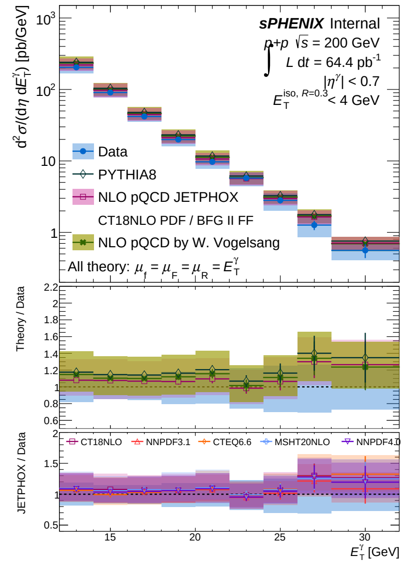
For each BDT partition variant we run `plot_final_selection.C` (the same macro used elsewhere for the nominal analysis and invoked by `make_selection_plots.sh`) to produce the two-panel cross-section + Theory/Data ratio figure. All five panels show the data-driven isolated-photon cross-section overlayed with PYTHIA 8, JETPHOX NLO (CT14 / BFG II FF, $\mu_f = \mu_F = \mu_R = E_T^\gamma$), and Vogelsang NLO; the luminosity label reads the matching analysis-config `lumi` value.



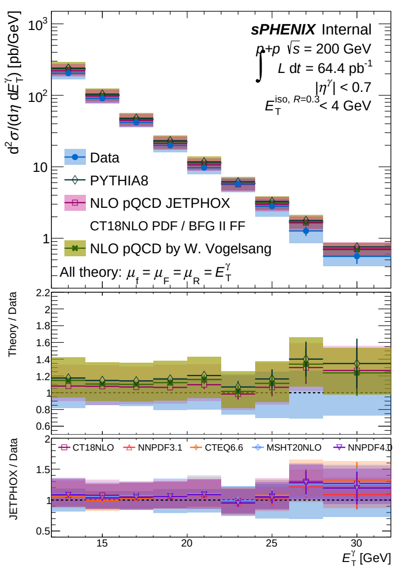
(a) parametric nominal (reference)



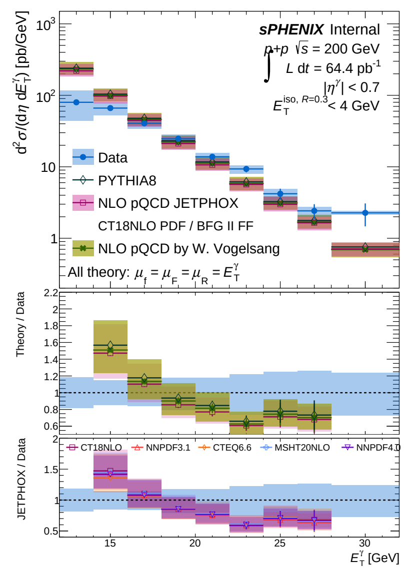
(b) `flat_t85_nt50`



(c) `flat_t90_nt50` (primary ask)



(d) `flat_t90_nt70`



(e) `flat_t95_nt50`

Figure 7: Data-driven isolated-photon differential cross-section $d^2\sigma/d\eta\, dE_T^\gamma$ for the five BDT-partition scan points, each extracted via the full pipeline (`RecoEffCalculator` on data + `CalculatePhotonYield` on data) with the signal-leakage-corrected ABCD purity and Bayesian-iterative unfolding. Bottom sub-panel of each figure shows Theory/Data with JETPHOX and Vogelsang NLO systematic bands.

10 Reviewer Pass Matrix

Table 5: Reviewer pass matrix for the artifacts produced in this scan.

Artifact	2a physics	2b cosmetics	2c re-derivation
`plot_purity_non-closure_ntbdt.C`	PASS (2 warnings on jet-MC labeling, now corrected; stale old files flagged, not used)	PASS (2 fixes: `strleg4` position, bottom-pad y-range)	PASS
`figures/purity_non-closure_ntbdt.pdf`	N/A	PASS	N/A
Numbers in Tables 1–2	N/A	N/A	PASS (independent re-extraction from ROOT matches)
`plot_purity_non-closure_flat.C`	PASS (per-variant truth-purity handling correct; four variants distinguished in region A)	PASS (1 iteration: fixed y-title offset, moved ISO label, widened left margin)	PASS (48 new rows appended to the scan CSV with `config_type=flat`)
`figures/purity_non-closure_flat_scan.pdf`	N/A	PASS	N/A
Numbers in Tables 3–4	N/A	N/A	PASS (independent re-extraction from ROOT matches)